

Supplement Recommendation Engine

Business User Guideline

A practical guide for clinicians, care coordinators, and pilot stakeholders
on what the solution does and how to use it day to day.

Audience: Business & clinical users | **Console:** <http://localhost:3000> | **Version:** Pilot-ready Docker stack (no real-time warehouse ingest)

Document type: Business User Guide · Confidential — internal pilot use

1. What this solution is

The Supplement Recommendation Engine is a **clinical decision support** system. It turns a patient profile (demographics, conditions, medications, labs, lifestyle) into **evidence-ranked nutraceutical recommendations**, each checked by a **deterministic safety gate** before anything is shown to a clinician.

It is designed to assist clinicians — not to replace professional judgement. Sessions that need human review are clearly flagged. Every recommendation includes a three-layer explanation (**Why · Evidence · Safety**) so the reasoning is transparent.

1.1 What you can do today

- **Score a stored pilot patient** from the cohort list and receive ranked recommendations.
- **Build an inline (dev) profile** with age, sex, region, BMI, conditions, meds, and labs.
- **Watch the safety pipeline** as the five gates run during scoring.
- **Review recommendation cards** — dose, confidence, UL usage, grade, gate checks.
- **Open Why / Evidence / Safety** tabs for each nutrient recommendation.
- **Approve, adjust, or reject** each recommendation (clinician feedback is stored).
- **Reload past sessions** from patient history when available.
- **Print or copy** a session summary for charts or handoff notes.
- **Toggle light / dark theme** for comfortable reading.

1.2 What is intentionally out of this pilot build

- Real-time bulk ingest from a hospital warehouse or LIS (planned for a later feeder project).
- Live Kafka event streaming (optional; not required to run the Docker pilot).
- FHIR HL7 parser and SMART-on-FHIR EHR launch (future phases).
- Automatic model retrain from clinician feedback (feedback is stored; night jobs are not).

2. Product features (business view)

2.1 Clinician console

The console at <http://localhost:3000> is the primary interface. It is a thin, web-based panel over the scoring API.



Figure 1 — Typical clinician workflow in the console

Feature	Business value
Stored patient intake	Pick a known pilot Patient UUID; engine loads profile from Postgres.
Inline profile (dev)	Build a one-off case without waiting for feeder data.
Safety pipeline animation	Makes the five deterministic gates visible during the run.
Evidence-ranked cards	Highest-priority nutrients first, with dose and confidence.
Gate chips / gate list	Shows Evidence, Interactions, Upper limit, Clinician gate status.
Why · Evidence · Safety	Transparent rationale for sign-off and teaching.
Clinician review banner	Stops silent release of high-risk doses / low confidence.
Approve / Adjust / Reject	Captures human override for audit and future quality loops.
Session history	Reopen prior scores for the same patient.
Print / Copy summary	Share results without giving API access to everyone.
Suppressed list	Shows nutrients blocked by the safety gate and why.

2.2 Safety & trust guarantees (non-technical)

- **Upper limit (UL) enforcement** — doses are capped; soft policy around 70–80% of UL.
- **Drug–nutrient interactions** — contraindicated pairs are blocked; majors are flagged.
- **Disease contraindications** — e.g. hemochromatosis + iron, advanced CKD + potassium.
- **Clinician escalation** — listed triggers force *requires clinician review*.
- **Immutable session trail** — each run is stored with model version and evidence snapshot.
- **Wellness disclaimer** — product output is decision support, not a prescribing system of record.

2.3 Example outcomes you can expect

Pilot style profile	Typical outcome you should see
T2DM + metformin (long duration)	B12 often ranked; interaction / demand signals in Why.
Low vitamin D lab + low sun / veiled	Vitamin D3 with dose vs UL meters; often Grade C/B band.
GERD + PPI (omeprazole)	Magnesium / B12 demand signals may appear.
Hemochromatosis seed	Iron must not appear in active recommendations.
Pregnancy seed	Guideline-aligned folate / iron consideration with safety text.

3. How to use the clinician console

3.1 Start the system (once)

With Docker Desktop running, from the **supplement_engine** folder:

```
python scripts/run_app.py up --open
```

This starts API, databases, knowledge graph, Redis, Nginx, and the console; seeds Neo4j evidence and 16 pilot patients; then opens the browser.

3.2 Score a stored patient (recommended pilot path)

- Open **http://localhost:3000**.
- Keep **Stored patient** selected.
- Choose a patient from the **Pilot cohort** dropdown (or paste a Patient UUID).
- Click **Run recommendation engine**.
- Watch the Safety pipeline complete, then read the result cards on the right.
- For each card: check dose, confidence, UL %, gate status; open Why / Evidence / Safety.
- Use **Approve**, **Adjust dose**, or **Reject** as your clinical decision.
- Optional: **Print summary** or **Copy summary** for documentation.

3.3 Score an inline (dev) profile

Switch to **Inline profile (dev)**, edit demographics / conditions / medications / labs, then run the engine. Use this for what-if teaching cases. Production pilots usually lock this mode off and only accept stored **patient_id** scoring.

3.4 Reading a recommendation card

Area	Meaning
Rank + name + form	Priority among nutrients that cleared the engine threshold.
Dose panel	Suggested amount, unit, frequency, food timing, and whether capped.
Evidence grade	A–D: guideline → meta-analytic → limited → mechanistic.
Confidence meter	Model confidence; low Confidence often forces clinician review.
Dose vs UL	How close the dose sits to the upper limit; red/amber near policy line.
Gates	Pass/flag for evidence, interactions, UL, clinician gate.
Tabs	Why (risk drivers), Evidence (strength), Safety (warnings / holds).
Footer actions	Approve / Adjust / Reject — commits clinician feedback to the audit trail.

4. Roles & responsibilities

Role	Uses the product to...
Clinician / nutritionist	Score patients, review rationales, approve or hold plans.
Clinical lead / pilot owner	Sign off persona checklist; confirm iron block & escalation cases.
Care coordinator	Run stored patients; print summaries for charting.
Compliance / QA	Recreate a session via session ID; review audit & evidence snapshot IDs.
IT / platform (light)	Start/stop Docker; confirm console & API reachability.

5. Safety disclaimer for users

Output is **wellness / clinical decision support** only. It is **not** a substitute for medical care, diagnosis, or a pharmacy order entry system. Always apply local protocols, lab context, and clinician judgement — especially when the console shows **requires clinician review**.

6. Quick links

Resource	URL / command
Clinician console	http://localhost:3000
API docs (Swagger)	http://localhost/docs
API health	http://localhost:8000/health
Start stack	<code>python scripts/run_app.py up --open</code>
Stop stack	<code>python scripts/run_app.py down</code>
Stack status	<code>python scripts/run_app.py status</code>

For deep technical architecture, APIs, databases, logging, and maintenance, see the companion PDF: **Supplement_Engine_Technical_Handover.pdf**.